

- 6 A cylindrical copper wire P of length 0.24 m is shown in Fig. 6.1.

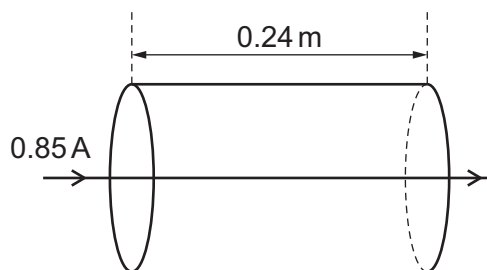


Fig. 6.1 (not to scale)

The current in the wire is 0.85 A.

The resistance of the wire is $3.3 \text{ m}\Omega$.

The **total** number of charge carriers N in the wire is 2.6×10^{22} .

The resistivity of copper is $1.8 \times 10^{-8} \Omega \text{ m}$.

- (a) Calculate the potential difference between the two ends of the wire.

potential difference = V [2]

- (b) (i) Show that the cross-sectional area of the wire is $1.3 \times 10^{-6} \text{ m}^2$.

[2]

- (ii) Show that the number density of charge carriers in the wire is $8.3 \times 10^{28} \text{ m}^{-3}$.

[1]

- (iii) Calculate the average drift speed of the charge carriers (electrons) in the wire.

average drift speed = ms^{-1} [2]





- (c) A different copper wire Q has the same volume as wire P, but non-uniform radius, as shown in Fig. 6.2.

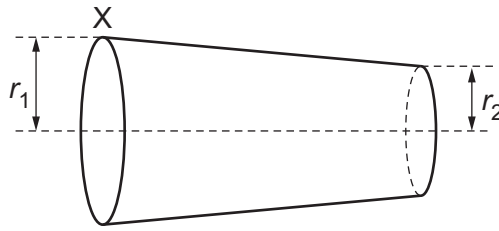


Fig. 6.2 (not to scale)

The radius r_1 at end X of wire Q is the same as the radius of wire P. Radius r_2 is less than r_1 .

- (i) State and explain how the resistance of wire Q compares with the resistance of wire P.

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..... [4]



- (ii) On Fig. 6.3, sketch a graph of the variation of the average drift speed of the charge carriers with distance from end X of wire Q.

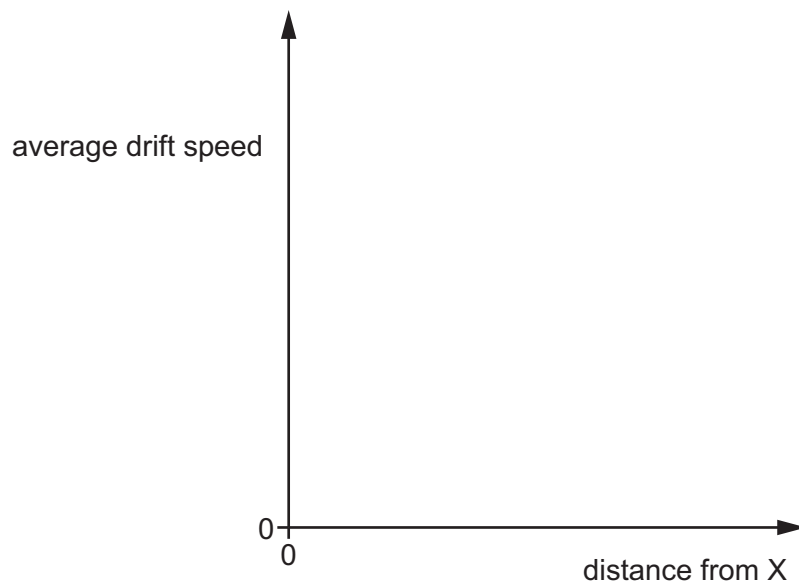


Fig. 6.3

[2]

[Total: 13]

